

AMINA EMAD

Cybersecurity & AI Student | SOC Analyst Aspirant | Network Security | Threat Detection
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PROFESSIONAL SUMMARY

Detail-oriented Artificial Intelligence student with a dedicated focus on Cybersecurity, Network Security, and Security Operations. Proficient in Wireshark for network traffic analysis and packet inspection, Nmap for active reconnaissance and port scanning, and Linux system administration. Skilled in Python security scripting and log analysis automation — directly applicable to SOC workflows, incident triage, and SIEM operations. Hands-on experience with IoT security architecture, biometric access control systems, and AI-driven anomaly detection. Seeking a Cybersecurity Internship or SOC Analyst Intern role to contribute to vulnerability management, threat monitoring, and real-world security event analysis.

TECHNICAL SKILLS

Security Tools & Concepts: Wireshark, Nmap, Network Traffic Analysis, Packet Inspection, Port Scanning, Cryptography (AES/SHA/RSA), Access Control, Authentication Systems, IDS/IPS Concepts, Vulnerability Assessment

Networking & OS: TCP/IP, OSI Model, Network Protocols, Linux (CLI & Administration), IoT Security, Embedded Network Security, Kali Linux (Familiar)

AI & Machine Learning: TensorFlow, PyTorch, Keras, Scikit-learn, OpenCV, Pandas, NumPy, Anomaly Detection, Pattern Recognition, Computer Vision

Programming & Scripting: Python (Security Scripting & Automation), C++, SQL, JavaScript (Basic)

Platforms & Tools: Git/GitHub, Linux CLI, Jupyter Notebook, Docker (Basic), Raspberry Pi, Arduino, ESP32, SIEM Concepts

EDUCATION

Bachelor of Artificial Intelligence (In Progress)

Expected: 2027

Faculty of Computers and Artificial Intelligence

Relevant Coursework: Network Security, Cryptography, Deep Learning, Computer Vision, Embedded Systems, Data Structures & Algorithms

PROJECTS

Biometric Access Control System | Python · OpenCV · Computer Vision

- Built a real-time facial authentication system using Python and OpenCV, implementing biometric identity verification to demonstrate access control security principles applicable to physical security monitoring.
- Classified authorized vs. unauthorized access attempts using ML pattern recognition algorithms, directly transferable to endpoint security monitoring and user authentication workflows in enterprise environments.

IoT Security Monitoring System | Raspberry Pi · ESP32 · Arduino · Embedded Systems

- Architected an IoT security platform integrating Raspberry Pi, ESP32, and multi-sensor inputs with automated event-driven alerts, demonstrating IoT attack surface awareness and embedded security control design.
- Designed real-time anomaly detection and incident-triggered alert mechanisms, mirroring SOC security event logging, alerting pipelines, and incident detection and response workflows.

Smart City AI Security Platform | Team Project · Machine Learning · Network Security

- Collaborated on an AI-driven urban security platform combining ML-based threat detection with IoT network sensors, addressing smart infrastructure cybersecurity vulnerabilities at scale.
- Developed network anomaly detection models simulating real-time threat monitoring workflows aligned with Security Operations Center (SOC) practices and security event management.

Neural Network Anomaly Classifier | Python · TensorFlow · Keras · Deep Learning

- Designed and trained a deep neural network from scratch for high-accuracy pattern classification, applying ML techniques directly relevant to AI-driven threat detection and behavioral anomaly identification in security operations.

RESEARCH EXPERIENCE

Co-Author — AI Security Research Paper | IEEE-Affiliated Conference Submission

- Co-authored and submitted a research paper to an IEEE-affiliated venue; led technical literature review, security-focused analysis, and academic writing — demonstrating rigorous research methodology and professional technical communication skills.

CERTIFICATIONS

Introduction to Cybersecurity – Cisco Networking Academy | Machine Learning Specialization – Coursera (Andrew Ng)

AWS Cloud Practitioner Essentials – Amazon Web Services | Python for Everybody – University of Michigan (Coursera)

LANGUAGES

Arabic (Native) **English** (Upper Intermediate – Professional Communication)